

The Effects of Capital Requirements on Good and Bad Risk-Taking

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We study capital requirement regulation in a dynamic quantitative model in which nonfinancial firms, as well as households, hold deposits. A novel general equilibrium channel that operates through firms deposits mitigates the cost of increasing capital requirements. In the calibrated model, (a) the optimal capital requirement is 7.3 percentage points higher than in a comparable model in which all the deposits are held by households, and (b) setting the capital requirement higher than the true optimum is not as costly as one would gauge from the comparable model. We also provide some independent evidence that supports our novel channel. (*JEL* E20, G21, G32)

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After the 2007–2008 financial crisis, the revision of the regulatory framework of financial intermediaries has become a central topic of discussion by regulators and academics. The Basel III accord has tightened bank regulation with the aim of reducing the likelihood and depth of financial crises. One of the key sets of rules at the center of this debate is capital requirements, namely, limits on the fraction of debt that banks can use to finance their investment. Appealing to the Modigliani and Miller (1958) theorem, Admati and Hellwig (2013) argue

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that capital requirements should be raised even further to eliminate banks' bad risk-taking, namely, the moral hazard induced by government guarantees and implicit too-big-to-fail subsidies.¹ One argument against tighter capital requirements is that banks are special because their liabilities are valued for their safety and liquidity (DeAngelo and Stulz 2015). Under this view, raising capital requirements reduces excessive risky lending by banks but at the cost of reducing the supply of valuable bank deposits.

Recent literature has analyzed these trade-offs in a quantitative, general equilibrium framework. While most analyses use models in which all the deposits are held by households (Van den Heuvel 2008; Davydiuk 2017; Begenau 2020), deposits in practice are held not only by households but also by firms. As of 2010, the year in which Basel III was agreed on, U.S. firms accounted for about one-third of the deposits held by households and firms.

In this paper, we ask whether and to what extent accounting for the deposits held by nonfinancial firms affects the determination of the optimal capital requirement. We derive a quantitative, general equilibrium model that we calibrate to U.S. data. Similar to the literature, government guarantees induce banks to take excessive risks. Capital requirements limit this risk, but they also reduce the supply of deposits. Crucially, though, the cost of reducing the supply of deposits depends not only on how households value deposits but also on how the lower supply affects firms' behavior.

Our main results are that (a) the optimal capital requirement is higher than in comparable models in which all the deposits are held by households, and (b) setting the capital requirement higher than the true optimum is not as costly as one would gauge from the comparable models. In our calibrated model, the optimal capital requirement is 21.9%, and only 14.6% in a comparable model in which households hold all the deposits. With respect to item (b), setting capital requirements 10 percentage points above the optimum erases almost all the welfare gains in the model in which all the deposits are held by households. But in the full model in which some of the deposits are held by firms, setting the capital requirement 10 percentage points above the optimal value delivers almost 90% of the welfare gains that are achieved at the optimum.

In our model, firms are subject to idiosyncratic shocks that lower their demand for production inputs—particularly labor—relative to an economy without such shocks (Quadrini 2017). Because deposits are safe, they implicitly provide insurance by reducing the volatility of firms' cash flows, so that a high availability of deposits increases firms' willingness to engage in risky production and hire workers.² We call this channel *good risk-taking* because

¹ Some regulators have made a case for similar rules as well (see, e.g., the Minneapolis Plan discussed by Kashkari 2016).

² The role of deposits as providers of insurance against idiosyncratic risk is consistent with the finding of Bates, Kahle, and Stulz (2009) of a rising trend in firms' cash holdings since the 1980s, which they attribute in large part to an increasing precautionary motive.

increasing labor demand is socially valuable, but not fully exploited because of the idiosyncratic risk.³ Deposits are also held by households, who derive utility from them as in Van den Heuvel (2008).

Our main result follows from the fact that, from the regulator's point of view, the benefit of deposits that arises from their safety and liquidity differs depending on whether deposits are held by households or firms: for every dollar of deposits, the value is much higher when the owner is a household. As a result, an increase in capital requirements that reduces the supply of deposits is not that detrimental for welfare in our model—in which some deposits are held by firms—in comparison to a model with the same total level of deposits which are all held by households.

Deposits held by firms are less valuable from the regulator's point of view because of a general equilibrium effect that works through firms' input prices, and in particular, wages. Tightening capital requirements reduces deposits and, thus, firms' labor demand. At constant wages, the reduction in labor demand would lead to a large decline in production and welfare. However, if labor supply is relatively inelastic, the equilibrium quantities of labor and production only fall by a little, and instead falling wages absorb most of the drop in demand. The lower wages do not affect welfare because households are both workers and firms' shareholders, and the reduction in wages translates into higher firms' profits. Thus, the aggregate costs of reducing liquidity supply in this model are lower than in a comparable one with the same level of deposits, but in which households hold all deposits and bear the entire cost of the policy change.

We also provide reduced-form evidence supporting the good risk-taking channel (i.e., the mechanism by which a higher availability of deposits affects employment and wages) using two different approaches that build on Jayaratne and Strahan (1996) and Drechsler, Savov, and Schnabl (2017), respectively. These analyses complement the evidence in Benmelech, Bergman, and Seru (2021), who show that firms' employment decisions are significantly affected by firms' liquidity.

Another contribution of the paper is the way we model the link between capital requirements and banks' bad risk-taking. To match the large spikes in banks' default rates in practice, we allow banks in our model to engage in endogenous and socially inefficient risk-shifting (Jensen and Meckling, 1976), which can be curbed by tighter capital requirements.⁴ Our formulation allows us to distinguish between total and risk-weighted assets, so that capital requirements are imposed on the latter, as is the case in practice. In addition, it matches the fact that an increase in the riskiness of banks' assets (measured by

³ Dindo, Modena, and Pelizzon (2020) derive a demand for banks' liabilities for insurance purposes similar to Quadrini (2017) and ours (although in a continuous-time model), but focus on the implications for business cycles and risk-free rates.

⁴ A large literature in banking focuses on the link between risk-shifting and competition (see, e.g., Repullo 2004; Boyd and De Nicolò 2005).

the ratio of risk-weighted to total assets) is followed on average by an increase in banks' default rates 1-year ahead.

Other aspects of our work build on the existing literature that quantifies optimal capital requirements. We follow the pioneering approach of Van den Heuvel (2008) in embedding the analysis of capital requirements regulation in a dynamic, quantitative, general equilibrium model, and some elements of our analysis are based on Van den Heuvel (2008), Davydiuk (2017), and Begenau (2020).⁵

Capital requirements in our model limit the negative impact of subsidized deposit insurance. More generally, though, deposit insurance can also be interpreted as any explicit or implicit government guarantee on bank debt (e.g., the Temporary Liquidity Guarantee Program set up by the Federal Deposit Insurance Corporation in 2008 and the too-big-to-fail guarantee that resulted in the Paulson plan studied by Veronesi and Zingales 2010). We follow a common approach in the literature that imposes deposit insurance and motivates it with its role in preventing runs, as in Diamond and Dybvig (1983), but does not explicitly include runs or analyze the optimality of such a policy.⁶ We note that government guarantees and capital requirements are important not only for households' but also for firms' deposits. Indeed, even if some firms might have high deposit balances, the government has been covering deposits and other bank liabilities well beyond the deposit insurance limit.⁷

We motivate firms' aversion to idiosyncratic risk with an agency problem, following a literature that has highlighted the importance of these frictions (Panousi and Papanikolaou 2012; Glover and Levine 2015; Glover and Levine 2017). Firms are owned by households that can fully diversify away their exposure to idiosyncratic risk by holding a well-diversified portfolio of equity. However, each firm is run by a manager who holds only an indifferentiated stake in the firm she manages and thus is exposed to the idiosyncratic risk of the firm. We introduce these features building on Quadrini (2017) and, as an additional contribution of the paper, we use some modeling assumptions that make the model tractable without altering the implications of the agency friction. In particular, managers' consumption does not affect the welfare analysis, and the firms' building block of the model can be aggregated easily. That is, even if idiosyncratic shocks create heterogeneity across firms, the equilibrium in our

⁵ We abstract from some features studied in other papers, such as the interaction with the shadow banking system (Begenau and Landvoigt 2021, Dempsey 2017), time-varying capital requirements (Davydiuk 2017, Malherbe 2020), the effect of legacy assets (Bahaj et al. 2016, Bahaj and Malherbe 2020), and the interaction with economic growth (Nguyễn 2014).

⁶ We note that fully eliminating deposit insurance might not be optimal in our framework because it would make deposits unsafe, thereby increasing rather than reducing the volatility of firms' cash flows. We leave the joint determination of optimal deposit insurance and capital requirement regulation to future research.

⁷ For instance, Veronesi and Zingales (2010) calculate that Paulson's equity infusion plan implemented during the Columbus Day weekend of October 2008—at the height of the financial crisis—entailed a transfer of \$21 billion to \$44 billion from taxpayers to the holders of liabilities of some of the largest U.S. banks, and this figure is likely a lower bound on the overall taxpayer-subsidized transfers that benefited all depositors.

model depends on aggregate firm wealth only, and other moments of the firm size distribution are irrelevant. This allows us to maintain the focus on financial regulation and to solve the model in general equilibrium. This last feature is particularly important for our novel channel; richer structural corporate finance models with agency frictions, such as Nikolov and Whited (2014), are instead typically solved in partial equilibrium.

1. Literature Review

A recent and growing literature has embedded the analysis of capital requirements into quantitative general equilibrium models. This literature includes Van den Heuvel (2008), Christiano and Ikeda (2013), Nguyễn (2014), Martinez-Miera and Suarez (2014), Gertler, Kiyotaki, and Prestipino (2016), Begenau and Landvoigt (2021), Davydiuk (2017), Dempsey (2017), Begenau (2020), and Elenev, Landvoigt, and Van Nieuwerburgh (2021).⁸ Most of the literature, however, tends to be qualitative or to abstract from general equilibrium (Thakor 2014). Indeed, several papers, such as Berger and Bouwman (2013), Mehran and Thakor (2011), and Diamond and Rajan (2001), provide results about the role of bank capital and changes in capital requirements that are identified in the cross-section, abstracting from general equilibrium effects.

A related paper by Allen et al. (2018) studies the effect of government guarantees on banks' choices of liquidity and investments in risky projects. Both our paper and that of Allen et al. (2018) point out that higher risk-taking can be a good consequence of financial regulation. In their paper, more risk-taking by banks is associated with greater liquidity provision. In our paper, more risk-taking by firms has a positive impact, in general, on employment and output.

This paper is also related to a broader banking literature that conducts policy analyses in models in which only households hold safe deposits, such as Stein (2012). While those theoretical results are not affected by our novel channel, the corresponding quantitative policy stance might need to be tilted by imposing stricter limits on intermediaries.

Our paper relates to a large literature on firms' exposure to idiosyncratic risks. Smith and Stulz (1985) and Froot, Scharfstein, and Stein (1993) derive theories in which firms do not completely hedge away idiosyncratic risks because external financing is costly. Minton and Schrand (1999) provide evidence that firms that are more exposed to idiosyncratic risk invest less and have higher costs of external financing than other firms, consistent with an inability or unwillingness to completely insure against idiosyncratic shocks.

⁸ Another strand of the literature uses quantitative industrial organization models to study imperfect competition in the deposit and loans markets and its interaction with financial regulation (see, e.g., Corbae and D'Erasmus 2021; Egan, Hortaçsu, and Matvos 2017).

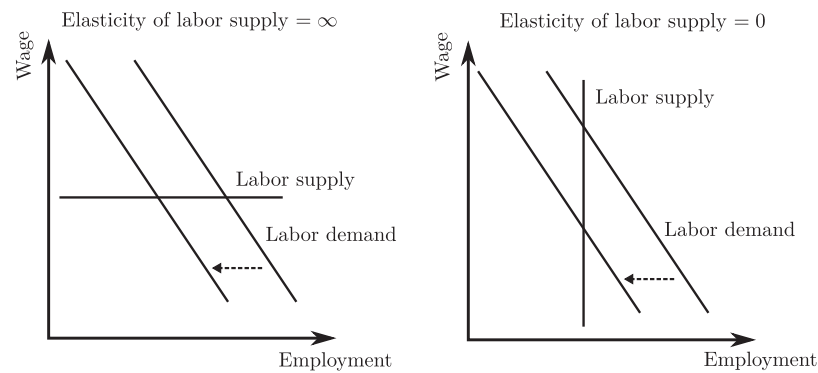


Figure 1
Equilibrium in the labor market and effects of a change in labor demand
Left panel, infinitely elastic labor supply; right panel, fully inelastic labor supply.

Panousi and Papanikolaou (2012) show that risk aversion of managers and their exposure to firms' idiosyncratic risk plays a role in shaping firms' decisions.

2. Intuition for Our Main Channel

In this section, we provide intuition behind the novel model mechanism that delivers our main result: relative to a model in which all the bank deposits are held by households, a model in which a fraction of the deposits are held by firms leads to a higher optimal capital requirement. We clarify our argument by breaking it down in several simple steps and analyzing each one in detail.

Consider first a baseline model (such as Van den Heuvel 2008) in which deposits enter the utility function of the representative household. In the literature, the parameters that govern households' utility from deposits are typically calibrated to match the observed liquidity premium on deposits, which is the difference between the return on deposits and the return on less liquid but still riskless assets. Second, consider an alternative model that includes a new class of deposit-holding agents (i.e., firms) but in which the total amount of deposits in the economy and the observed deposit premium are the same. In this new model, only a fraction (say, $X\%$) of the observed deposit premium is due to the household's direct utility for deposits; the remaining fraction ($100-X\%$) comes from firms' demand for deposits. In our model, firms use deposits to lessen the negative impact of uninsurable idiosyncratic shocks. In particular, a higher availability of deposits reduces the volatility of firms' assets and cash-flows. This, in turn, encourages firms to increase their production and, in particular, hire more workers.

Next, consider the effects of raising capital requirements, which reduces the quantity of deposits, in both models. In the first model, in which households hold 100% of deposits, the reduction in deposit is fully absorbed by households and reduces their utility, which ultimately lowers welfare. As shown by

Van den Heuvel (2008), the welfare effect depends on the liquidity premium (and a few other observables), and is higher the higher the premium is to start with. In the second model, in which some of the deposits are held by firms, however, the reduction in deposits is absorbed in part by households and in part by firms. With lower deposits, firms reduce their labor demand according to the good risk-taking channel explained before. The key insight is that while a reduction in deposit supply reduces firms' demand for labor, what matters for welfare is what happens to the *equilibrium* quantity of labor, which depends also on labor *supply*. In particular, the *elasticity* of labor supply crucially affects the outcome. Figure 1 clarifies this result by considering two extreme cases.

The left panel of Figure 1 depicts a completely elastic labor supply. In this case, wages are not affected by a change in demand, and a reduction in labor supply leads to a one-for-one reduction in employment, which reduces welfare, in line with the outcome of the model with only households' deposits.

The right panel of Figure 1 depicts the other extreme case: a fully rigid labor supply (i.e., the elasticity of labor supply is zero). A reduction in labor demand does not affect employment, and instead only lowers wages. However, the reduction in wages leads to higher firms' profits, and the two effects cancel out as long as workers are also the owners of firms—which will be the case in our model. As a result, the $100 - X\%$ of the liquidity premium due to firms' deposits produces no welfare consequences through the labor market.⁹ The logic of Van den Heuvel (2008) still applies to the $X\%$ of the deposit premium which is due to households' deposits. But the welfare costs of higher capital requirements are lower than in the case in which 100% of the deposit premium is due to households' deposits. As a result, the optimal capital requirement is higher. Notice also that the fall in wages need not be large—if firms' labor demand is very elastic, a modest change in wages is sufficient to keep employment constant.

In sum, the extent to which firm deposits affect welfare thus depends on the Frisch elasticity of labor supply. The recent micro- and macro-labor literature suggests that the Frisch elasticity is quite low, and we use similar values in our calibrated model. Hence, the welfare effects of raising capital requirements that arise through the labor market are modest. Our estimates are also conservative in that we assume a high degree of frictions in the labor market that prevent wages from responding quickly to shocks.

Absent from this discussion, by design, is any consideration of the benefits of increased capital requirements, because we take them to be the same in the two models. We model these benefits as arising from a reduction in

⁹ When firms use not only labor but also capital, a reduction in firms' deposits brings about some welfare consequences even if labor supply is inelastic. With fewer deposits, firms would reduce not only the demand for labor but also that for capital. Importantly, since capital in RBC-type models like those used in the capital requirement literature has an infinite supply elasticity, the equilibrium level of capital would change. Our quantitative analysis includes these additional welfare effects, because firms in our model use both labor and capital.

banks' risk-taking behavior at higher capital requirements (what we call "bad" risk-taking). Taken together, we argue that a higher capital requirement in an economy with a somewhat-inelastic labor supply will reduce banks' bad risk-taking—which is good for welfare—with only a limited impact on employment, because falling wages absorb most of the drop in labor demand. As a result, a regulator can set a much higher capital requirement, relative to a model in which 100% of the deposit premium comes from households' utility for deposits. The model that we will present next shows that this argument is quantitatively relevant in practice.

3. Quantitative Model

We now present a dynamic general equilibrium model that we then use to conduct a quantitative analysis and to determine the optimal capital requirement. Time is discrete and infinite. There are four players in the economy: firms, banks, households, and the government.

We follow Quadrini (2017) in assuming that firms are subject to idiosyncratic risk and hold deposits for precautionary reasons, but we depart significantly from his model. In particular, each firm is run by a risk-averse manager who acts in her own interest because of an agency friction. This friction motivates the manager's holding of an undifferentiated stake in the firm she runs and, thus, an exposure to firm-specific idiosyncratic risk (Panousi and Papanikolaou 2012; Glover and Levine 2015; Glover and Levine 2017). In contrast, firms' owners (i.e., households) fully diversify away the idiosyncratic risk by holding equity stakes in all firms. The firms' side of the model aggregates up to a "representative firm"—even though firms are heterogeneous because of the idiosyncratic shocks—keeping the model tractable.

Banks collect deposits from firms and households, provide loans to firms, and are subject to idiosyncratic shocks that make a fraction of them insolvent. Deposits at insolvent banks are made whole by the government through deposit insurance.

Households consume, supply labor and bank equity, and own firms and banks. We will impose some parameter restrictions so that the objective of financial regulation will be solely that of maximizing the welfare of households, without any consideration for the welfare of managers.

3.1 Firms and managers

The economy is populated by a continuum of firms owned by households. Each firm is run by a manager who behaves in her own interest because of an agency friction. The manager's compensation depends on a contract that includes a fixed component, an equity stake, and an option.¹⁰ The manager

¹⁰ The contract between the firm and the manager is imposed exogenously. Our approach is to use a compensation scheme that we will later discipline from the data, and to derive implications for firms' behavior and capital requirements.

makes all the firms' choices other than the decision about dividends, which is instead controlled by the shareholders (i.e., the households). We return to the motivation for this assumption when presenting the household side of the model, in Section 3.3.

Each firm i starts the period with wealth a_t^i that the manager allocates between deposits d_t^i and firms' direct holdings of productive assets k_t^i (i.e., physical capital)

$$d_t^i + k_t^i \leq a_t^i, \quad (1)$$

Deposits pay, at $t+1$, a gross return R_t^d and are safe because the government provides full deposit insurance.

The manager makes two additional choices at time t : one-period loans b_t^i from banks with interest rate r_t , and labor l_t^i with wage w_t . Loans b_t^i obtained from banks are chosen subject to a borrowing constraint

$$b_t^i \leq \xi k_t^i, \quad (2)$$

where $0 < \xi < 1$. Bank loans are used by the firm to get additional productive assets, so that the total capital used in production is the sum of owned capital and capital borrowed from banks: $k_t^i + b_t^i$.¹¹

The production function of firm i is

$$y_t^i = A_{t+1} z_{t+1}^i (k_t^i + b_t^i)^\gamma (l_t^i)^{1-\gamma}, \quad (3)$$

where A_{t+1} is the aggregate technology and z_{t+1}^i is a firm-specific idiosyncratic productivity shock. While our results hold for a general specification of z_{t+1}^i , we focus on the case in which $z_{t+1}^i \in \{z^L, z^H\}$ with probability $1 - p_z$ and p_z , respectively, and $E\{z_{t+1}^i\}$ is normalized to one.

Both aggregate productivity A_{t+1} and the idiosyncratic shock z_{t+1}^i are realized at $t+1$, that is, after the manager has made the time- t choices described above. In other words, the interest on the loan $r_t b_t^i$ and the wage bill $w_t l_t^i$ are not contingent on the realization of A_{t+1} or z_{t+1}^i .¹² In the event of negative productivity shocks, the manager can use the deposits plus their return, $R_t^d d_t^i$, to compensate for any shortfall in cash flow and pay the promised amounts to workers and banks.

At the beginning of $t+1$, the total wealth available to the firm is

$$x_{t+1}^i = y_t^i - w_t l_t^i - r_t b_t^i + R_t^d d_t^i + (1 - \delta) z_{t+1}^i k_t^i, \quad (4)$$

where y_t^i is output defined in (3), $w_t l_t^i$ is the wage bill, $r_t b_t^i$ is the interest paid on bank loans, $R_t^d d_t^i$ denotes the deposits plus the return, and $(1 - \delta) z_{t+1}^i k_t^i$ is

¹¹ We can equivalently write the borrowing constraint as $b_t^i \leq \hat{\xi}(k_t^i + b_t^i)$, where $\hat{\xi}$ has the interpretation of the pledgeability of the total capital $k_t^i + b_t^i$ used by the firm. The constraint can be rearranged as $b_t^i \leq k_t^i \hat{\xi} / (1 + \hat{\xi})$ and thus is equivalent to our formulation as long as $\xi = \hat{\xi} / (1 + \hat{\xi})$.

¹² Noncontingency on z_{t+1}^i follows Quadrini (2017) and is based on the assumption that the z_{t+1}^i are uninsurable. In principle, we could make the interest rate on bank loans and the wage contingent on A_{t+1} , but the formulation we use improves the fit with the untargeted moments.

the undepreciated capital. The stock of capital k_t^i is subject to the same shock z_{t+1}^i that affects productivity, thereby making the firms' assets risky and leaving deposits as the only safe asset in which firms can invest.¹³

The wealth x_{t+1}^i is then split between resources that are retained inside the firms (i.e., next period's assets a_{t+1}^i) and those that are paid out (i.e., to finance the manager compensation c_{t+1}^i and dividends π_{t+1}^i). We denote α_{t+1}^i to be the fraction of wealth to be paid out, which is chosen by shareholders (see Section 3.3), so that

$$\pi_{t+1}^i + c_{t+1}^i \leq \alpha_{t+1}^i x_{t+1}^i. \quad (5)$$

The fraction $1 - \alpha_{t+1}^i$ of the wealth x_{t+1}^i is retained inside the firm, but the wealth a_{t+1}^i is also affected by dividend adjustment costs as in Jermann and Quadrini (2012):¹⁴

$$a_{t+1}^i = (1 - \alpha_{t+1}^i) x_{t+1}^i - \frac{\varphi}{2} x_{t+1}^i (\alpha_{t+1}^i - \bar{\alpha})^2, \quad (6)$$

where $\varphi \geq 0$ parametrizes the dividend adjustment cost. The formulation in (6), in which the dividend adjustment cost depends on the deviation of α_{t+1}^i from its steady-state level $\bar{\alpha}$, allows us to maintain the tractability of the firms' side of the model.

The compensation c_{t+1}^i that the manager receives at $t+1$ depends on three elements: (1) a fixed component proportional to firms' size a_t^i at time t , but not contingent on the manager's choices or the firm's cash flow at $t+1$; (2) an equity stake; and (3) an option component. While the results hold qualitatively if we include only the equity stake—which is the most important source of compensation in the data—we also allow for the other components to produce more accurate quantitative results. Total compensation is given by

$$c_{t+1}^i = \kappa [\theta_f a_t^i + \theta_e \alpha_{t+1}^i x_{t+1}^i + \theta_o \max\{0, a_{t+1}^i - a_t^i\}], \quad (7)$$

where κ scales the entire level of the compensation package and θ_f , θ_e , and θ_o parameterize the importance of the fixed, equity, and option components, respectively. Equation (7) implicitly assumes that the payoff of the option, if exercised, is equal to the growth of the firm's assets, $a_{t+1}^i - a_t^i$. Internet Appendix C shows that, in our parameterization, the payoff $a_{t+1}^i - a_t^i$ is equal to that of an option issued at the money, as in Glover and Levine (2017).¹⁵

¹³ In Section 4.5, we consider an extension to allow firms to hold additional safe assets with the same return R_t^d . Our main results are essentially unchanged so long as the supply of these additional safe asset does not vary with changes in capital requirements.

¹⁴ Dividend adjustment costs allow the model to match the volatility of firms' deposits; without them, firm deposit volatility in the model is counterfactually low.

¹⁵ The max operator of the option payoff gives rise to a kink in the the manager's objective function. To keep the analysis tractable, we approximate the option compensation so that such payoff is awarded if and only if the firm is hit by the idiosyncratic shock $z_{t+1}^i = z^H$ (see Internet Appendix C for more details). In our benchmark calibration, we find that the approximation holds exactly for every simulated point.

We can now formalize the problem of the manager. The manager solves

$$V_t^m(a_t^i) = \max_{d_t^i, k_t^i, l_t^i, b_t^i, c_{t+1}^i, a_{t+1}^i} \beta^m E_t \left\{ \frac{(c_{t+1}^i)^{1-\gamma_m}}{1-\gamma_m} + V_{t+1}^m(a_{t+1}^i) \right\}, \quad (8)$$

subject to the firm budget constraint (1), the borrowing constraint (2), the law of motion of the firm's wealth (6), and the compensation contract (7), and where $\beta^m \in (0, 1)$ is the discount factor of the manager and γ_m is her risk aversion.

The manager's choices expose her consumption to risk, which she dislikes because of the concavity of the utility function. In principle, the manager can choose to have little or no exposure to risk by investing most of a_t^i in deposits, getting no loans from banks, and hiring few workers. Such choices would make her consumption c_{t+1}^i deterministic. However, the manager will find it optimal to make her consumption risky by investing in capital, getting bank loans, and hiring workers. This allows her to benefit from the higher return of firms' production relative to deposits.

We can now characterize the optimal choice of the manager. The problem boils down to choosing the firm's capital k_t^i and labor l_t^i . Given k_t^i and l_t^i , deposits are determined by the budget constraint (1), bank loans are determined by the borrowing constraint (2), and firms' assets at $t+1$ and manager consumption at $t+1$ are given by (6) and (7), respectively.

The next proposition formalizes the manager's choices, focusing on the case in which the borrowing constraint (2) is always binding. We then verify numerically, when simulating the model, that this is the relevant case. The constraint is binding because banks fund part of their balance sheet with deposits—which are cheap as they include a liquidity premium—and thus are able to offer a rate that is lower than firms' own funding costs. As a result, firms want to max out on what they borrow from banks. Internet Appendix A contains all the proofs.

Proposition 1. (Manager's choices) If the borrowing constraint (2) is binding and the variance of aggregate risk is sufficiently small relative to that of idiosyncratic risk (see Internet Appendix C.2), manager i 's optimal choice of firm's own capital, labor, and borrowing from banks are

$$k_t^i = \phi_t^k a_t^i, \quad l_t^i = \phi_t^l a_t^i, \quad b_t^i = \xi k_t^i,$$

where ϕ_t^k and ϕ_t^l are independent of a_t^i and satisfy the following first-order conditions:

$$\mathbb{E}_t \left\{ \left[\gamma A_{t+1} z_{t+1}^i (1+\xi)^\gamma \left(\frac{\phi_t^l}{\phi_t^k} \right)^{1-\gamma} + z_{t+1}^i (1-\delta) - \xi r_t - R_t^d \right] \Lambda_{t+1}^m \right\} = 0 \quad (9)$$

$$\mathbb{E}_t \left\{ \left[(1-\gamma) A_{t+1} z_{t+1}^i (1+\xi)^\gamma \left(\frac{\phi_t^k}{\phi_t^l} \right)^\gamma - w_t \right] \Lambda_{t+1}^m \right\} = 0 \quad (10)$$

and Λ_{t+1}^m is the marginal utility of consumption of the manager, defined in Internet Appendix A.

The first-order conditions (9) and (10) for capital and labor have a simple asset pricing interpretation: the excess returns from investing in capital and hiring additional workers are zero at the optimum, when discounting using the manager's marginal utility Λ_{t+1}^m . They also imply that the firms' side of the model can be described by a single "representative firm." This is because the first-order conditions are independent of a_t^i . As a result, even if the size of individual firms grows and shrinks as they receive different sequences of idiosyncratic shocks, we can easily solve for the equilibrium because the firm-size distribution does not affect aggregate variables. Given the total amount of firms' assets a_t^f ,

$$a_t^f = \int a_t^i di, \quad (11)$$

the following corollary shows that the behavior of the firm sector can be described by a representative firm. We state the corollary for the case in which all firms follow the same dividend policy, that is, $\alpha_{t+1}^i = \alpha_{t+1}$ for all i . This is indeed the outcome in equilibrium, as shown in Section 3.3.

Corollary 1. Given a_t^f , the total capital held by firms $k_t = \int k_t^i di$, the total labor demand $l_t = \int l_t^i di$, and the total amount borrowed from banks $b_t = \int b_t^i di$ are given by

$$k_t = \phi_t^k a_t^f, \quad l_t = \phi_t^l a_t^f, \quad b_t = \xi k_t, \quad (12)$$

where ϕ_t^k and ϕ_t^l are independent of the distribution of $\{a_t^i\}$ and are determined by the first-order conditions in (9) and (10). In addition, in the relevant case in which $\alpha_{t+1}^i = \alpha_{t+1}$, the law of motion of total firms' asset is

$$a_{t+1}^f = \left[1 - \alpha_{t+1} - \frac{\varphi}{2} (\alpha_{t+1} - \bar{\alpha})^2 \right] R_{t+1}^x a_t^f \quad (13)$$

where R_{t+1}^x is the cum-dividend return of a dollar invested in the firm sector:

$$R_{t+1}^x \equiv A_{t+1} [\phi_t^k (1 + \xi)]^\gamma (\phi_t^l)^{1-\gamma} - w_t \phi_t^l - r_t \xi \phi_t^k + (1 - \delta) \phi_t^k + R_t^d (1 - \phi_t^k). \quad (14)$$

Dividends paid to shareholders are $\pi_{t+1}^f = \alpha_{t+1} R_{t+1}^x a_t^f - \int c_{t+1}^i di$.

Another useful implication of Proposition 1 is that the first-order conditions of the manager are independent of the parameter κ that scales the level of the compensation package of the manager (Equation (7)). We will exploit this feature when solving the model and performing welfare calculations. In particular, we will impose parametric restrictions that imply that the total consumption of managers is small relative to the overall consumption in the economy—formally, κ will be assumed to be arbitrarily close to zero—while at the same time not affecting the implications of the agency friction.

Corollary 2. The results of Proposition 1 are independent of κ .

The results of Proposition 1 imply that our model is consistent with some relevant evidence. First, as in Opler et al. (1999), (a) riskier cash flow (i.e., higher variance of z_{t+1}^i) implies that the firm is willing to hold more cash, and (b) firms that do well (i.e., firms hit by a sequence of good values of z_{t+1}^i) accumulate cash internally, and vice versa. Second, as firms accumulate cash, they increase their labor force, in line with the evidence of Benmelech, Bergman, and Seru (2021) about the effect of cash held by the firm on the firm's labor force. Third, lower cash flow volatility increases investments, along the line of the evidence in Minton and Schrand (1999), and because capital and labor are complementary in production, increases labor demand too.

3.2 Banks

Banks live for a single period: each bank is set up at time t and is liquidated at the beginning of time $t + 1$.¹⁶ Each newly created bank receives an amount n_t of net worth from its shareholders (i.e., households). Then, the bank collects deposits d_t from firms and households, subject to a capital requirement constraint, and uses the resources $n_t + d_t$ to purchase physical capital b_t , which represent loans to firms. Profits are bounded below by a limited liability constraint, and when a bank fails, it is taken over and liquidated by the government. Deposits are protected and fully repaid because of the deposit insurance.

Banks are subject to idiosyncratic shocks that affect their cash flow at $t + 1$. These shocks capture the fact that some banks experience higher or lower losses on their loans and investments in comparison to the average bank in the economy. We assume a baseline level of exogenous risk that banks cannot get rid of, so that some banks will always fail each period.

In addition to the baseline level of idiosyncratic risk, banks can engage in risk shifting (Jensen and Meckling 1976; Repullo 2004) by increasing the volatility of their idiosyncratic shock, at a cost.¹⁷ We add the risk-shifting element to the model for quantitative reasons—without, it the model fails to reproduce the large spikes in banks' default rates during crisis periods, although our main qualitative results continue to hold as we show in Section 4.5. Other papers use a similar approach to improve the quantitative performance of their models. For instance, in Begenau (2020), banks can choose a suboptimal level of monitoring to increase the bailout subsidy they receive from the government.

To clarify the mechanics of the model, we note that limited liability and deposit insurance imply that banks have an incentive to increase their exposure to idiosyncratic risk. They can do so in two ways: (1) by increasing the size of their balance sheet and (2) by engaging in risk shifting. We call this behavior bad risk-taking because although it is privately optimal for banks, it is socially

¹⁶ This assumption is made without loss of generality because our model does not include adjustment costs on banks' size nor costs to raise equity.

¹⁷ The cost reduces the expected return of banks' investments when banks engage in risk-shifting, as is normally assumed in the risk shifting literature.

wasteful: banks reap the profits if their realized idiosyncratic shock is high, but pass the costs on to the government if the realization is low. Two related remarks are important. First, risk shifting creates value for individual banks but is socially costly because engaging in it comes at a real cost. Hence, the socially optimal amount of risk shifting is zero. Second, even without risk shifting, banks can take advantage of the deposit insurance subsidy by expanding their balance sheet beyond what would be socially optimal.

To describe the problem of banks, consider first the case in which banks do not to engage in risk shifting. Each bank lends its physical capital b_t to the firms, and after production takes place at $t+1$, firms return the undepreciated fraction $1-\delta$ plus a return r_t . The resources $b_t[1-\delta+r_t]$ returned by firms are then hit by the idiosyncratic shock ε , with $E(\varepsilon)=1$. Thus, banks' profits at $t+1$ are given by the cash flow $\varepsilon b_t[1-\delta+r_t]$ net of the repayment $R_t^d d_t$ to depositors, where R_t^d is the gross return on deposits.

The full problem of banks includes the choice of risk shifting, which we denote by $p_t \geq 0$, in addition to that of loans b_t and deposits d_t :

$$\max_{b_t, d_t, p_t \geq 0} E_t \int_{\underline{\varepsilon}_t}^{\infty} \{ \varepsilon b_t [1 - \delta + r_t - \lambda(p_t) r_t] - R_t^d d_t \} dF(\varepsilon; p_t), \quad (15)$$

subject to the budget and capital requirement constraints:

$$b_t = d_t + n_t, \quad (16)$$

$$n_t \geq \zeta \frac{b_t}{1 - \nu p_t}, \quad (17)$$

where ζ is the capital requirement, $\nu \geq 0$ parameterizes the dependence of the capital requirement constraint on the risk-shifting choice p_t (discussed below), and $\underline{\varepsilon}_t$ is the threshold of the idiosyncratic shock ε below which a bank defaults.¹⁸

The risk-shifting choice p_t has two effects on the bank's profits. First, p_t affects the distribution of the idiosyncratic shock $F(\varepsilon; p_t)$. In particular, we assume that the variance of ε is increasing in p_t but the mean is not affected, that is, $E(\varepsilon)=1$ for all p_t . With higher variance, banks' default rate goes up, increasing the reliance on deposit insurance and, as a result, the deposit insurance subsidy. This effect increases expected bank profits. Second, a choice $p_t > 0$ is costly, in the sense that it reduces the return on bank profits by $\lambda(p_t)r_t$. Banks trade off these two effects when setting p_t to maximize their expected profits. We assume $\lambda(0)=0$ and $\lambda'(p) > 0$, and we specify the functional forms of $F(\cdot)$ and $\lambda(\cdot)$ when calibrating the model, in Section 4.1. In particular, under our parameterization, the risk-shifting constraint $p_t \geq 0$ will be occasionally binding. As a result, the model will produce "normal times" with low default

¹⁸ The threshold is implicitly defined by $R_t^d d_t = \underline{\varepsilon}_t b_t [1 - \delta + r_t - \lambda(p_t) r_t]$.

rates (i.e., $p_t = 0$) and “crisis times” with spikes in the default rate (i.e., $p_t > 0$), as in the data.

In addition to bank’s profits, the risk-shifting choice p_t affects the capital requirement constraint (17): as banks engage in riskier activities, their capital requirement constraint tightens. This effect mirrors how banks are regulated in practice: starting with Basel I, capital requirements are applied to risk-weighted assets as opposed to total assets. Beyond risk-weighted capital requirements, banks are subject to additional regulatory scrutiny, such as inspections and CAMELS ratings, which also affect the riskiness of their loans (Rezende and Wu 2013; Gopalan, Kalda, and Manela 2021; Hirtle, Kovner, and Plosser 2020). We model the link between p_t , risk-weighted capital, and the capital requirement with the parameter ν , which we discipline in the calibration by interpreting the term $b_t/(1 - \nu p_t)$ on the right-hand side of (17) as risk-weighted assets.

Accounting for the endogenous reaction of regulation to bad risk-taking is necessary to produce an accurate policy analysis. For example, a regulatory constraint that does not account for p_t , such as one of the form $n_t \geq \zeta b_t$, would imply that a reduction in ζ would lead banks to massively increase p_t , with very negative consequences on welfare. In practice, however, risk-weights, inspections, and CAMELS ratings attenuate this effect, and our formulation captures this outcome. In other words, omitting p_t from (17) would overestimate the welfare impact of changing capital requirements.

The maximization problem (15) requires solving for the size of the banking sector, b_t , and the amount of risk shifting p_t . Here, we discuss solving for b_t , and we postpone the discussion of p_t to Section 4.1, after we specify the functional forms for $F(\cdot)$ and $\lambda(\cdot)$.

To solve for the banks’ size, we note that the capital requirement constraint (17) is always binding in equilibrium because the return on deposits will be less than the return on banks’ capital. Given n_t (and p_t), Equation (17) pins down the size b_t of banks’ assets, and then the budget constraint (16) yields the value of deposits d_t . Finally, the equilibrium value of n_t is determined by households (described in Section 3.3), given the return on equity R_{t+1}^n . Because households can diversify their holdings across banks, Equation (15) implies that R_{t+1}^n is given by

$$R_{t+1}^n = \frac{1}{n_t} \int_{\varepsilon_t}^{\infty} \{ \varepsilon b_t [1 - \delta + r_t - \lambda(p_t) r_t] - R_t^d d_t \} dF(\varepsilon; p_t). \quad (18)$$

3.3 Households

Households are infinitely lived agents that enjoy utility from consumption c_t and disutility of labor l_t . In addition, we follow Van den Heuvel (2008), Begenau (2020), and several other papers in the capital requirements literature by assuming that households enjoy utility from holding deposits d_t^h , a feature that captures the liquidity and transaction benefits that are provided by bank

deposits above and beyond their pecuniary return.¹⁹ As mentioned before, households are owners of both firms and banks and thus receive firms' dividends and the return on bank equity.

Households' utility is given by

$$\sum_{t=0}^{\infty} \beta^t \left[\frac{c_t^{1-\gamma_m} - 1}{1-\gamma_m} + \psi \frac{(d_t^h/c_t)^{1-\gamma_d}}{1-\gamma_d} - \beta \chi \frac{l_t^{1+\frac{1}{\eta}}}{1+\frac{1}{\eta}} \right], \quad (19)$$

where $\beta \in (0, 1)$ is the subjective discount factor, $\gamma_m > 0$ is the coefficient of relative risk aversion, $\gamma_d > 0$ and $\psi \geq 0$ govern the demand of deposits, and $\chi > 0$ and $\eta > 0$ govern the disutility of labor (and thus, labor supply). Households choose labor at time t but do not provide that labor until the beginning of period $t+1$, when production actually occurs; thus, we discount the disutility of labor chosen at t with an additional β in Equation (19). We postpone the discussion of labor supply and the labor market to Section 3.4, so that this section illustrates the other households' choices taking l_t as given.

Households choose consumption, investments in bank equity, deposits, and—in their role as firms' shareholders—the firms' dividend policy (i.e., the fraction α_t^i of firms' wealth that is paid out). The assumption that dividends are chosen by households—as opposed to managers—is motivated by the results of La Porta et al. (2000). They find that in countries with good legal protections for shareholders, such as the United States, firms choose dividend policies that are consistent with shareholders' rather than managers' preferences.²⁰

A household that starts with wealth a_t^h solves the problem

$$V_t^h(a_t^h) = \max_{c_t, n_t, d_t^h, \{\alpha_t^i\}_i} \frac{c_t^{1-\gamma_m} - 1}{1-\gamma_m} + \psi \frac{(d_t^h/c_t)^{1-\gamma_d}}{1-\gamma_d} - \beta \chi \frac{l_t^{1+\frac{1}{\eta}}}{1+\frac{1}{\eta}} + \beta E_t \{ V_{t+1}^h(a_{t+1}^h) \}, \quad (20)$$

subject to the budget constraint

$$c_t + n_t + d_t^h \leq a_t^h. \quad (21)$$

At time t , the household chooses how to allocate its wealth a_t^h between consumption c_t , deposits d_t^h , and investment in bank equity n_t . Wealth at $t+1$ is given by

$$a_{t+1}^h = w_t l_t + n_t R_{t+1}^n + d_t^h R_{t+1}^d + \pi_{t+1}^f - T_{t+1}. \quad (22)$$

That is, a_{t+1}^h is the sum of labor income $w_t l_t$, the gross return on banks' equity $n_t R_{t+1}^n$ and deposits $d_t^h R_{t+1}^d$, and the profits π_{t+1}^f distributed by firms, net of lump-sum taxes T_{t+1} .

¹⁹ In principle, one can think about the effect of deposits in providing insurance against idiosyncratic risk to households, as we do for firms. However, we abstract from this channel here and leave it for future research.

²⁰ In common-law countries, even minority shareholders enjoy a high level of legal protection and can easily monitor firms' cash. In addition, shareholders can substantially influence dividend payments. For instance, for public firms, activist investors are likely to take actions if cash and dividends are managed in a way that hurts shareholders. To prevent such actions, managers might follow policies that are most beneficial for shareholders in the first place even if they have some control over the dividend policies.

In addition to consumption and investment, households also choose the dividend policy of firms, α_t^i for all i (see Section 3.1). They do so to maximize their valuation of each firm, balancing the benefits of leaving cash in the firm—which provides insurance to managers' exposure to idiosyncratic risk and thus increases good risk-taking—with the value of an additional dollar of dividends.

The next proposition summarizes the household's optimal choices of consumption, investment, and firms' dividends, focusing on the relevant case that we will analyze when solving the model in which the parameter κ that governs the manager's compensation is small.

Proposition 2. Denoting Λ_t to be the households' marginal utility of consumption, the households' optimality conditions for the choice of bank equity n_t and deposits d_t^h are

$$1 = E_t \left\{ \beta \frac{\Lambda_{t+1}}{\Lambda_t} R_{t+1}^n \right\}, \quad (23)$$

$$1 = E_t \left\{ \beta \frac{\Lambda_{t+1}}{\Lambda_t} \right\} R_t^d + \psi \frac{1}{\Lambda_t} c_t^{-1} \left(\frac{d_t^h}{c_t} \right)^{-\gamma_d}. \quad (24)$$

The optimal dividend policy, when $\kappa \rightarrow 0$, implies that $\alpha_t^i = \alpha_t$ for all i and is chosen so that

$$1 = E_t \left\{ \beta \left(\frac{\Lambda_{t+1}}{\Lambda_t} \right) R_{t+1}^x [1 + \varphi(\alpha_t - \bar{\alpha})] \right\}, \quad (25)$$

where R_{t+1}^x is the cum-dividend return of a dollar invested in the firm sector, defined in (14).

Two key results are noticeable from Proposition 2. First, Equations (23), (24), and (25) are the standard asset pricing conditions for the return on equity R_{t+1}^n , the return on deposits R_t^d , and the return on firms' assets R_{t+1}^x . With respect to Equation (24), the second term on the right-hand-side reflects the liquidity benefits generated by deposits. With respect to Equation (25), the term $\varphi(\alpha_t - \bar{\alpha})$ accounts for the effects of the dividend adjustment costs. Second, the dividend policy is the same for all firms, that is, $\alpha_t^i = \alpha_t$ for all i , in line with the result of Section 3.1 that the firms' side of the model can be represented by the behavior of a representative firm.

In addition, the optimal dividend policy in Equation (25) implies that the model generates a deposit premium that is linked to firms' production and wages. This is easier to see in the nonstochastic steady state, in which (25) can be rewritten as

$$\frac{1}{\beta} - R^d = R^l \underbrace{\frac{\partial l^i}{\partial a^i}}_{=\phi^l} + R^k \underbrace{\frac{\partial k^i}{\partial a^i}}_{=\phi^k} > 0, \quad (26)$$

where we have dropped the time subscript because we are focusing on the steady state, and where R^l and R^k are the excess returns on additional labor and capital, respectively, and averaged across firms:

$$R^l \equiv (1 - \gamma)A(1 + \xi)^\gamma (\phi^k)^\gamma (\phi^l)^{-\gamma} - w,$$

$$R^k \equiv \gamma A(1 + \xi)^\gamma (\phi^k)^{\gamma-1} (\phi^l)^{1-\gamma} + 1 - \delta - r\xi - R^d.$$

To understand (26), consider the cost and benefit of reducing firms' dividends so that each firm in the economy holds an additional dollar. The cost, on the left-hand side, is given by the deposit premium (i.e., the fact that the return R^d on deposits is less than the risk-free illiquid return, which is $1/\beta$ in the nonstochastic steady state). The benefit is captured by the terms on the right-hand side, which capture the excess return from the additional unit of labor and capital that firms use when they hold more deposits.

Crucially, the right-hand side of (26) represents the *private* benefit of firms' holdings of deposits because it is computed by taking prices as given—in particular, the wage. In contrast, a regulator that wants to set the optimal capital requirement in the model considers the *social* benefit, which accounts for general equilibrium changes in wages. As argued in Section 2 and quantified in Section 4, the change in the wage leads to a higher optimal capital requirement in comparison to a model in which all the deposits are held by households.

3.4 Labor market

The labor market is competitive, but wages are rigid in the sense that they adjust slowly over time in response to shocks. Our formulation for wage rigidities mirrors those derived in the macro-labor literature that microfound such rigidities using infrequent bargaining between workers and firms (Gertler and Trigari 2009; Gertler, Huckfeldt, and Trigari 2020).

Let w_t^f be the frictionless level of the wage that would prevail without wage rigidities. We assume that the actual wage w_t is given by

$$w_t = \omega w_{t-1} + (1 - \omega) w_t^f, \quad (27)$$

where $\omega \in [0, 1]$ parametrizes the degree of wage rigidity. The model nests a standard case with fully flexible wages (i.e., $\omega = 0$) as well as a case in which the wage rigidity is extreme (i.e., $\omega \rightarrow 1$). Given w_t , the amount of labor l_t is determined by firms' demand, similar to Shimer (2012). The frictionless wage w_t^f is defined to be the one that would satisfy the first-order condition of the households in a frictionless model:

$$w_t^f E_t \{ \Lambda_{t+1} \} = \chi l_t^{1/\eta}. \quad (28)$$

Equation (28) is similar to a standard labor supply condition, with the difference that it depends on the marginal utility of consumption at $t + 1$, (i.e., Λ_{t+1}) rather than at t . This is because, in our model, firms hire workers at time t but pay the wages at $t + 1$.

Because our main channel operates through the adjustment of wages following a change in capital requirements, the wage rigidities in Equation (27) dampen our main channel and make the analysis more conservative. In particular, the short-term dynamics of the wage w_t are crucially affected by the rigidity parameter ω : at the time of a change in capital requirements ζ , a high value of ω implies that wages do not absorb as much of the shock, leading to a larger fall in labor—and thus lower welfare—than would obtain in a frictionless model. This dampening effect of ω thus works against any inelasticity in labor supply coming from a low value of the Frisch elasticity η , as discussed in Section 2. On the other hand, wage rigidities do not play any role in steady-state (i.e., the long run), in which equilibrium in the labor market is determined by the interaction of firms' labor demand and households' labor supply—in line with the modeling approach used in the macro-labor literature.

3.5 Government

The government taxes households to ensure that depositors at failed banks are made whole. At $t+1$ the government seizes output at failed banks (who return zero to their equity holders) in order to partially defray the expenses of paying back depositors.

The total amount of tax to be collected is

$$T_{t+1} = \int_{-\infty}^{\varepsilon_t} [R_t^d d_t - \varepsilon b_t [1 - \delta + r_t - \lambda(p_t)r_t]] dF(\varepsilon; p_t), \quad (29)$$

where $R_t^d d_t$ is the amount owed to depositors, and the term $\varepsilon b_t [1 - \delta + r_t - \lambda(p_t)r_t]$ for $\varepsilon < \varepsilon_t$ denotes the assets of failed banks.

3.6 Aggregate risk

We assume that the productivity of firms, A_t , follows an AR(1) process in logarithms. The law of motion of A_t is

$$\log A_t = (1 - \rho) \log \bar{A} + \rho \log A_{t-1} + \sigma_A \varepsilon_t^A,$$

where $\bar{A}, \sigma_A > 0$ and $\varepsilon_t^A \sim \mathbb{N}(0, 1)$.

3.7 Equilibrium definition

Given an initial conditions for aggregate firm assets a_0^f , household wealth a_0^h , and the exogenous stochastic processes for A_t , $\{z_t^i\}$ and $\{\varepsilon\}$, an equilibrium is a collection of firm policies, bank policies, household policies, government taxes, wages, and return on deposits such that: managers maximize their expected utility (8); banks maximize their expected profits (15); households maximize their expected utility (19); the government budget constraint (29) holds; the wage w_t is determined by Equations (27) and (28); and the return on deposits R_t^d clears the deposit market.

The aggregate resource constraint will hold by Walras' law. Nonetheless, we state it to clarify the mechanics of the model:

$$c_t + (k_t + b_t) \leq (1 - \delta)(k_{t-1} + b_{t-1}) + A_t(k_{t-1} + b_{t-1})^\gamma l_{t-1}^{1-\gamma} - \lambda(p_{t-1})r_{t-1}b_{t-1} - \frac{\varphi}{2}x_t(\alpha_t - \bar{\alpha})^2.$$

The resources available to the economy, on the right-hand side, consist of the undepreciated capital plus output (recall that output produced by firms depends on choices made at $t - 1$) and net of the resources that are wasted because of risk shifting and the dividend adjustment costs.²¹ The left-hand side states that these resources can be consumed or invested in capital.

3.8 Welfare

When computing optimal capital requirements, we should, in principle, account for the welfare of both types of agents in the economy: households and managers. However, doing so requires taking a stand on the weight to be assigned to each type in the welfare function—a task that would significantly complicate the analysis. To address this issue, we note that the fraction of managers in the population is small relative to households, and their compensation is small relative to firms' profits. In this section, we formalize the notion that total consumption and total welfare are well approximated by households' consumption and welfare, respectively, so that we can compute the optimal capital requirement by looking solely at the welfare of households.

In general, the welfare function W is the sum of households' and managers' welfare:

$$W = V_0^h(a_0^h) + \theta \int V_0^m(a_0^i) di, \quad (30)$$

where $\theta > 0$ is a Pareto weight that defines managers' contribution to total welfare in comparison to households. The welfare analysis depends, in principle, on both the Pareto weight θ and the parameter κ that scales the compensation of managers. We discipline these two parameters in such a way that (a) welfare depends only on households' utility, $W = V_0^h(a_0)$; (b) aggregate consumption depends only on households' consumption; and (c) the implication of the agency friction and the easy aggregation of the firms' side of the model continue to hold.

To discipline θ , we note that the fraction of managers in the population is, in practice, small. As a result, we consider θ to be arbitrarily small so that the economywide welfare depends only on households' utility: $W = V_0^h(a_0)$. In Internet Appendix D, we provide a simple extension of the model in which we derive the result $W = V_0^h(a_0)$ even if we set the Pareto weight θ to one

²¹ In our quantitative analysis, we define GDP in the model as output net of the resources wasted because of risk shifting and the dividend adjustment cost; in other words, GDP is consumption plus investment.

(i.e., even if each manager has the same Pareto weight as each household). To discipline κ , we note that managers' compensation are small relative to the overall dividends paid by firms and thus we set $\kappa \approx 0$.²² We formalize these choices and their implications in the next proposition.

Proposition 3. (Parameter restrictions and welfare) If $\kappa \rightarrow 0$ and $\theta \rightarrow 0$ (with θ proportional to κ^{γ_m} along the limiting sequence), manager i 's optimal choices in Proposition 1 are not affected, manager i 's consumption converges to $c_{t+1}^i \rightarrow 0$, profits distributed to households become $\pi_{t+1}^f \rightarrow \alpha_{t+1} x_{t+1}$ and the economywide welfare function W becomes equal to the value function of households, $W \rightarrow V_0^h(a_0)$.

The key result of Proposition 3 is that the importance of managers vanishes for welfare purposes (i.e., their consumption converges to zero) without affecting their decisions. As a result, we can just focus on the welfare of households when evaluating financial regulation, even though the model exhibits a behavior driven by an agency friction.

4. Calibration and Quantitative Analysis

We now take the model to the data. To gauge the impact of our novel channel on the optimal capital requirement, we provide two calibrations: a benchmark calibration in which both households and firms hold deposits, and a restricted calibration in which all the deposits are held by households. More precisely, the restricted calibration features no idiosyncratic risk for firms, so that firms have no reason to hold deposits. We match both calibrations to the same target moments—the full calibration, since it has more parameters, also targets a few more moments—and compute the optimal capital requirement for each calibration.

We calibrate the model to annual U.S. data from 1986 to 2010. This choice maximizes the length of the data sample, while focusing on a period of time in which (a) banks were allowed to pay interest on most deposit instruments and (b) there was a single capital requirement regime in the sample. The majority of Regulation Q restrictions on interest rates were phased out between 1980 and 1985 (Gilbert 1986), and capital regulation throughout most of our sample was determined by Basel I, with the switch to the Basel III agreement and the Dodd-Frank Act signed in 2010.

We solve the model using nonlinear global projection methods that allow for rich dynamics and occasionally binding constraints (see Internet Appendix G for more details). In particular, the constraint that is occasionally binding is the

²² Setting an arbitrarily small κ drives managers' consumption to zero. However, we show in Internet Appendix D that when we formalize the measures of managers and take the limit as this measure goes to zero, managers' consumption remains positive and well defined even if $\kappa \rightarrow 0$.

one that determines banks' risk shifting, that is, $p_t \geq 0$ (see Equation 15). As discussed in Section 3.2, risk shifting allows banks to increase their exposure to idiosyncratic risk to exploit the implicit deposit insurance subsidy. This feature of the model allows us to match both the spikes in bank failures during crisis times and the long periods with few bank failures.

We begin by specifying the functional forms that govern banks' shocks and risk shifting (Section 4.1) and then turn to calibrating parameter values (Section 4.2). Starting with an 8% capital requirement, we simulate the model 2,500 times for 25 periods — so that the number of periods corresponds to the number of years in our sample — and then show how various untargeted moments in the model compare to the data (Section 4.3).²³ Finally, we study the effects of changing the capital requirement in the baseline calibration (Section 4.4) and under several alternative calibrations (Section 4.5).

4.1 Banks' bad risk-taking: Functional forms

Before calibrating the model, we specify the functional forms that govern banks' idiosyncratic shocks and risk-shifting decisions. Recall from Section 3.2 the two functions that we need to specify: the function $\lambda(p_t)$ that parameterizes the costs associated with the risk-shifting choice p_t , and the cumulative distribution function (CDF) $F(\varepsilon; p_t)$ that governs the banks' idiosyncratic shocks ε .

We begin by discussing the choice of the function $\lambda(p_t)$. In principle, we do not know the exact functional form, and making wrong assumptions could affect the policy analysis. However, we exploit the fact that any (well-behaved) function can be approximated arbitrarily well by a polynomial function (i.e., the Stone-Weierstrass approximation theorem). Thus, we can write

$$\lambda(p_t) \approx \sum_{j=0}^J \lambda_j p_t^j$$

for some $J \geq 0$. We choose $J=2$, so that we are taking a second-order Taylor approximation of the true $\lambda(p_t)$ function. Thus, we have

$$\lambda(p_t) = \lambda_0 + \lambda_1 p_t + \lambda_2 p_t^2.$$

We set $\lambda_0=0$, so that $\lambda(0)=0$ (i.e., there are no costs associated with choosing $p_t=0$), and we calibrate λ_1 and λ_2 in Section 4.2 to match some key moments of the default rate distribution in the data.

Next, we turn to the idiosyncratic shock ε and its CDF $F(\varepsilon, p_t)$. We assume that the idiosyncratic shock ε is

$$\varepsilon \sim \begin{cases} 0 & \text{with probability } p_t \\ \frac{\tilde{\varepsilon}}{1-p_t} & \text{with probability } 1-p_t, \end{cases} \quad (31)$$

where $\tilde{\varepsilon}$ is a lognormal random variable with variance σ and $E\{\tilde{\varepsilon}\}=1$. Note that (31) implies, by construction, that $E\{\varepsilon\}=1$ for all values of p_t . The need

²³ More precisely, we simulate the model 125 periods and discard the first 100 observations.

to have a tractable specification in our rich model plays a role in selecting this functional form, but we show in Table 2 in Section 4.3 that the model matches several untargeted moments related to banks' choices and default.

The risk-shifting choice p_t , when the constraint $p_t \geq 0$ is not binding, is determined by the first-order condition

$$\lambda'(p_t)r_t[1 - F^{\ln}(\underline{\varepsilon}_t)] = Pr\{\varepsilon \geq \underline{\varepsilon}_t\}R_t^d \left[1 - \zeta \frac{1 - \nu}{(1 - \nu p_t)^2}\right], \quad (32)$$

where $F^{\ln}(\cdot)$ denotes the CDF of the lognormal distribution. The marginal cost of increasing p_t depends on the direct effect of the $\lambda(\cdot)$ function, which reduces the expected return on loans (i.e., the term on the left-hand side of Equation (32)). The marginal benefits are related to the reduction in the expected cost of repaying the deposits that are internalized by the bank. As banks increase p_t , they are more likely to fail, thereby shifting the burden of repaying the return R_t^d toward the deposit insurer. The marginal benefits of shifting the failure costs to the deposit insurer are smaller as the capital requirement ζ increases, since a higher capital requirement implies that banks are funded with a smaller fraction of deposits.²⁴

4.2 Calibration

In this section, we describe how we choose parameter values for the full calibration (in which both firms and households hold deposits) and the restricted calibration (in which all the deposits are held by households). We divide the parameters into two groups: the first group, reported in panel A of Table 1, includes those that we normalize, set to values commonly employed in the literature, or based on independent evidence. We set the second group of parameters, reported in panel B of Table 1, to match target moments from the data.

4.2.1 Fixed parameters. Panel A of Table 1 reports the fixed parameters. We normalize the steady state level of productivity $\bar{A}$, the mean of the firms' idiosyncratic shocks $E\{z^i\}$, and the disutility of labor χ to one. For several parameters, we use standard values from the literature.²⁵ We set the discount factor β of households and β^m of managers to 0.95 (i.e., the risk-free rate is 5% in the nonstochastic steady state), the depreciation δ to 0.1, the autocorrelation of productivity shocks ρ to 0.95, and the coefficient γ of capital in the production function to 0.3. We set the risk aversion of managers γ_m to three, as in Glover and Levine (2015, 2017), and the households' risk aversion γ_c to three as well, which is in line with the values used in the literature (see, e.g.,

²⁴ This requires $\nu \leq 1$, which is satisfied in all our calibrations.

²⁵ See, for example, King and Rebelo (1999) and Jermann and Quadrini (2012).

Barro 2006). The steady-state value of the dividend policy $\bar{\alpha}$ follows from (13) and (25) evaluated in steady state, which imply $\bar{\alpha} = 1 - \beta$.

To discipline the parameters θ_f , θ_e , and θ_o that govern the manager compensation, we use executive compensation data from ExecuComp. Following Dittmann and Maug (2007) and Glover and Levine (2017), we measure (a) the fixed pay as the sum of salary, bonus, other compensations, and long-term pay incentives as a fraction of total assets; (b) the shares of equity owned by CEOs; and (c) the options awarded to CEOs as a fraction of common shares outstanding (see Internet Appendix B.1 for more details). Because the manager's choices in Proposition 1 depend on the relative importance of the fixed, equity, and option compensation only, we can normalize one of the parameters θ_f , θ_e , θ_o to one. We choose to set $\theta_e = 1$, so that the ExecuComp data imply $\theta_f = 0.0297$ and $\theta_o = 0.0960$.²⁶

We set the Frisch elasticity of labor supply η to one, which is the upper bound of the range suggested by Chetty et al. (2011). This choice is also in line with the recent macro-labor literature based on microdata (Dyrda, Kaplan, and Ríos-Rull 2012; Heathcote, Storesletten, and Violante 2010). We set the parameter γ_d that governs households' liquidity value from deposits to 1.4, which is the same value used by Davydiuk (2017) and Begenau (2020).

4.2.2 Parameters calibrated to match data moments. Panel B of Table 1 reports the parameters that we calibrate toward the indicated data moment. While each parameter affects all moments, panel B of Table 1 indicates target moments that are particularly affected by each parameter. We focus most of the discussion on the full calibration with both firms' and households' deposits, and comment briefly at the end about the restricted calibration in which all the deposits are held by households.

We begin by calibrating the parameters that determine the deposits held by households and firms. Households' and firms' demand for deposits are determined by ψ and the distribution of the idiosyncratic shocks z_t^i , respectively. As noted in Section 3.1, z_t^i can take values z^H , z^L with probabilities p_z and $1 - p_z$, respectively. Given p_z (which we calibrate to match another moment, as discussed below), we set $\psi = 0.00248$ and $z^L = 0.9024$ to match the fraction of deposits held by households vs. firms and the liquidity premium. To measure these two quantities in the data, we proceed as follows. For deposits, we use data from the Flow of Funds (FF) and the Survey of Consumer Finances (SCF).²⁷ We find that households hold about two-thirds of the total deposits held jointly by households and firms (see Internet Appendix B.2 for more details). We

²⁶ Recall from Section 3.1 that the overall level of the manager compensation is scaled by κ . As discussed in Section 3.8, we focus on the limiting case $\kappa \rightarrow 0$, so that total welfare is approximated with that of households without affecting managers' choices in Proposition 1.

²⁷ Firms' deposits are based on the FF, which accounts for deposits of privately held firms and, thus, might differ from results based on Compustat.

Table 1
Calibrated parameter values

A. Preset parameters

Parameter	Value	Parameter	Value	Parameter	Value
η	1	γ	0.3	$\bar{A}$	1
β, β^m	0.95	γ_c	3	$E\{z^i\}$	1
δ	0.1	ρ	0.95	χ	1
$\bar{\alpha}$	0.05	γ_m	3	γ_d	1.4
θ_e	1	θ_f	0.0297	θ_o	0.0960

B. Parameters chosen to match data moments

Parameter	Value		Moment target	Moment value	
	Full	HH only		Data & full	HH only
ψ	0.00248	0.00446	Deposit premium, $R^f - R^d$	1.12%	1.12%
ξ	0.3002	0.3054	Deposits-to-GDP ratio	0.43	0.43
σ	0.027	0.0275	Average bank default rate	0.61%	0.61%
λ_1	7.29281	7.42856	Low-default times	68.0%	68.0%
λ_2	2.68	3.22	p85 bank default rate	1.55%	1.55%
ω	0.623	0.619	Vol(employment)/vol(GDP)	0.58	0.58
ν	0.80	0.99	Vol(risk-weighted assets)/vol(assets)	1.27	1.27
σ_A	0.0142	0.01465	Volatility logGDP	1.75%	1.75%
z^L	0.90235	1	Fraction of households' deposits	66.4%	100%
p_z	0.838	1	Continuers employment growth	2.50%	—
φ	1.32	0	Vol(firm deposits)/vol(GDP)	3.49	—

In panel B, “Full” denotes the full calibration with both households’ and firms’ deposits, and “HH only” denotes the restricted calibration in which all the deposits are held by households. All volatilities are computed by taking the logarithm and applying the HP-filter with parameter 100, both in the model and in the data.

cross-check the total deposits from the FF and SCF with data on deposits from the FDIC and the National Credit Union Administration, and the two measures are consistent with each other. We measure the liquidity premium in the data as the difference between the rate on 3-month AA commercial paper, which proxies safe illiquid assets as in Davydiuk (2017), and the rate paid by banks on deposits. We measure the latter as the ratio of interest expense on domestic deposits relative to the stock of such deposits, using FDIC data. This approach yields a liquidity premium of 1.12% per year (see Internet Appendix B.3 for more details).

The size of the banking sector in the model is affected by the ratio of bank lending, b_t , to physical capital owned by the firms, k_t . This ratio is governed by the borrowing constraint (2) and, in particular, the borrowing limit ξ . We set $\xi = 0.3002$ to match the 0.43 deposits-to-GDP ratio in the data.

We set the parameter p_z that governs firms’ probability of high and low idiosyncratic shocks to match the average employment growth rate of firms that expand their labor force, corresponding to firms that receive the high value of z_{t+1}^i in the model. We define the average employment growth rate as

$$g_{t+1}^i \equiv \frac{\hat{l}_{t+1}^i - l_t^i}{\frac{1}{2}(\hat{l}_{t+1}^i + l_t^i)},$$

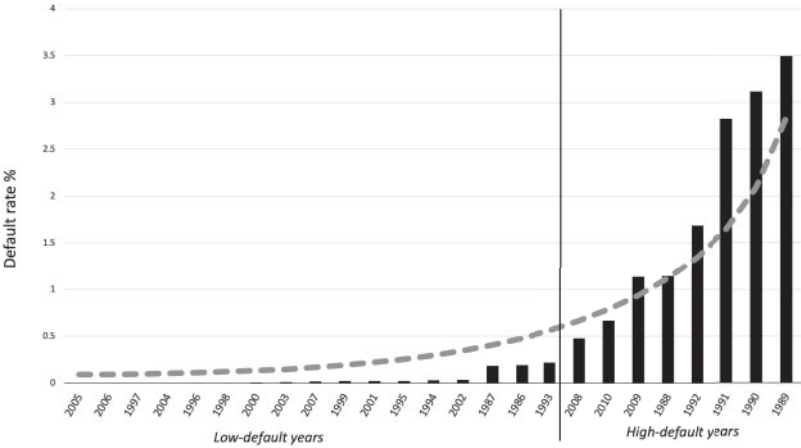


Figure 2

Bank default rates, sorted, in the data (bars) and model (dashed line)

Banks' default rate in the data is measured as the fraction of banks that are taken over by the FDIC (weighted by assets) and for which the FDIC incurred a positive resolution cost. In the model, we sort the default rate in each simulated path and then average it out across paths.

where $l_t^i = \phi_t^i a_t^i$ and $\hat{l}_{t+1}^i = \phi_{t+1}^i a_{t+1}^i \big|_{z_{t+1}^i = z_H}$. Given the structure of our model, the growth rate g_{t+1}^i is independent of a_t^i , so that the target does not depend on the distribution of firms' assets. Haltiwanger, Jarmin, and Miranda (2013) show that for all but the youngest firms, this growth rate is around 2.5% per year.

The parameters σ , λ_1 , and λ_2 govern banks' default probability, as discussed in Section 4.1. We discipline them using bank default rates, measured as the fraction of banks that are taken over by the FDIC (weighted by assets) and for which the FDIC incurred a positive resolution cost. The average default rate is 0.61% per year, in line with Elenev, Landvoigt, and Van Nieuwerburgh (2021), which we target by setting $\sigma = 0.027$. The parameter λ_1 determines whether the risk-shifting constraint $p_t \geq 0$ is binding or not—when it is binding (i.e., $p_t = 0$), banks do not risk-shift and the probability of default is low, and when it does not (i.e., $p_t > 0$), banks engage in risk shifting and the default rate spikes. Thus, we set λ_1 to match the fraction of low- versus high-default years, which we define as follows. We sort the yearly default rates in the data from the lowest to the highest. They are plotted in Figure 2 (bars), where we also include, for comparison, the same statistic from the model, taking averages across simulations (dashed line). We define high-default years to be (a) the 2008–2009 crisis and (b) all other years with a default rate higher than that. Item (b) implies that 1988–1992 (i.e., most of the Saving and Loan crisis) and 2010 are also high-default years.²⁸ Overall, 68% of the years in the sample are

²⁸ Including additional years associated with the Saving and Loan crisis to the high-default times (i.e., adding 1986, 1987, and 1993) would increase bad risk-taking in the model and make the optimal capital requirement higher.

defined as low-default times, and we set λ_1 so that, in the model, $p_t=0$ and default is low 68% of the time. Finally, because λ_2 governs bad risk-taking in the model in high-default times, we set it to 2.69 to match the 85th percentile of the default rate distribution, which corresponds approximately to the median default rate in high-default times.

To calibrate the parameter ν that governs the dependence of the capital requirement on the risk-shifting choice p_t , we use the fact that capital requirements in practice require banks to hold a minimum amount of equity as a fraction of risk-weighted assets. Hence, we interpret the term $b_t/(1-\nu p_t)$ on the right-hand side of (17) as risk-weighted assets. We then note that ν affects the volatility of risk-weighted assets $b_t/(1-\nu p_t)$ relative to the volatility of total assets b_t , and set ν to match the corresponding data moment. Empirically, we use FDIC data and consider only the bank assets that have a close correspondence to the model (i.e., those that finance physical capital; see Internet Appendix B.4). To compute risk-weighted assets in the data, we apply Basel I weights to each category of balance sheet item that we consider. We provide more details in Internet Appendix B.4, where we also compare our measure of risk-weighted assets with the one computed by the FDIC.

We set φ to match the volatility of the logarithm of firm's deposits, which is 3.49 times as high as the volatility of logarithm of gross domestic product (GDP). We set the standard deviation of the productivity process, σ_A , to match the volatility of the logarithm of GDP. We set the wage rigidity parameter ω to 0.623 to match the volatility of employment.

We obtain the restricted calibration in which all the deposits are held by households by shutting down the frictions in the production side of the model other than the need to have some of the physical capital financed by bank loans. As a result, our model is as close as possible to Begenau (2020) and other papers in the literature. In particular, (a) we shut down firms' idiosyncratic risk by setting $p_z = 1$, which implies that firms invest all their wealth a_t^i in capital and households hold 100% of the deposits; (b) we set $\varphi=0$ and select the remaining parameter values to match the same targets as in the full calibration; and (c) we replace managers' marginal utility, Λ_t^m , with that of households, Λ_t , to solve for firms' labor demand in (10), so that managers have no independent role in firms' decision-making.

4.3 Simulation and untargeted moments

Table 2 lists nontargeted moments for both the full and restricted models, though in what follows we focus most of the discussion on the full calibration with both firms' and households' deposits.

The model matches several nontargeted moments well. These include business cycle moments (e.g., the volatility of investments and wages, and the correlation of GDP with investments, consumption, and lagged GDP) as well as several moments related to banks and deposits (e.g., volatility of bank profits and its correlation with GDP, the correlation between the deposit premium

Table 2
Untargeted moments

Moment (untargeted)	Data	Full	HH only
Volatilities, ratios to log GDP, HP-filtered			
log(investment)	3.77	3.61	3.63
log(consumption)	0.69	0.41	0.41
log(wages)	0.77	0.76	0.77
log(firm deposits / labor)	3.27	2.94	–
Bank profits / GDP	0.17	0.18	0.14
log(1+deposit premium)	0.52	0.04	0.01
Correlations with log GDP, HP-filtered			
log(investment)	0.92	0.98	0.98
log(consumption)	0.90	0.89	0.89
log(wages)	0.40	0.86	0.87
log(firm deposits)	0.67	0.35	–
log(labor)	0.89	0.20	0.22
log(firm deposits / labor)	0.55	0.38	–
Bank profits / GDP	0.53	0.34	0.35
log(lagged GDP)	0.68	0.47	0.47
Other correlations, HP-filtered			
Corr(log[1+deposit premium], log[deposits])	–0.22	–0.30	0.21
Corr(log[1+deposit premium], log[labor])	0.82	0.95	0.70
Corr(log[1+deposit premium], log[wages])	–0.26	0.23	0.69
Corr(Δ log[deposits], log[labor]) ^a	0.76	0.33	0.33
Corr(Δ log[deposits], log[wages]) ^a	0.33	0.58	0.59
Corr(log[risk-weighted assets], bank default rate $t+1$) ^a	0.44	0.53	0.45
Corr(log[risk-weighted/total assets], bank default rate $t+1$) ^a	0.39	0.84	0.75
Other moments, not HP-filtered			
Vol. household deposits / GDP, every 3 years	2.98	2.10	0.98
Vol. bank default rate	1.06	0.76	0.64
FDIC resolution costs to assets of failed banks	19%	28%	27%
Corr(bank default rate, Δ log GDP)	–0.27	–0.30	–0.25
Banks' assets, % of capital stock	24.4	23.1	23.4

The moments in the first three panels are computed by applying the HP-filter both to the data and model-simulated series, using a smoothing parameter of 100. *a*: Δ log deposits and bank default rate are not HP-filtered. “Full” denotes the full calibration with both households’ and firms’ deposits, and “HH only” denotes the restricted calibration in which households hold all the deposits.

and deposits/labor, and the ratio of banks’ assets to the stock of capital). The fit is also good with respect to the volatility of households’ deposits, which is computed using one observation every 3 years because of data availability from the SCF.²⁹

We emphasize that the model matches well several moments related to the default rate of banks, which is a central element of the bad risk-taking side of the

²⁹ In the model, we compute this moment similarly by taking one observation out of every 3. Alternatively, one could compute data moments about households’ deposits using the Distributional Financial Accounts (DFA), which is available at higher frequency. An issue with this data is that the DFA overestimates the deposits truly held by households as recorded by the SCF, so the comparison with the model is only approximate. Nonetheless, the fit with the DFA is good. The volatility of log(household deposits) is 2.61 in the data and 2.08 in the model. The correlation with the logarithm of GDP is 0.27 in the data and –0.06 in the model, but the data moment is not statistically significant. Essentially, the link between GDP and household deposits is flat both in the model and the DFA data.

model. These include the correlation of bank default rates with GDP growth, the volatility of bank default, and the average FDIC resolution costs relative to the assets of failed banks. We also compute the correlation between risk-weighted assets and the one-period-ahead bank default rate. In the model, as banks increase risk-shifting $p_t > 0$, risk-weighted assets go up at time t and the default rate goes up at $t + 1$. This is the case in the data (correlation: 0.44), and the model matches this moment well. The fit is also good when we consider the ratio of risk-weighted to total assets—as opposed to just risk-weighted assets—which captures shifts toward riskier investments in the data for any given level of assets.

Although the model seems to miss a few untargeted moments, a careful inspection reveals that the model nonetheless performs relatively well. The volatility of the deposit premium is substantially lower than in the data, but this result is common in macro-banking models; see for instance Davydiuk (2017). The correlation of firms' deposits and labor with GDP are somewhat lower than in the data; however, when we consider the ratio of firms' deposits to labor—motivated by the link between these two quantities in the main channel that affects our results—the data fit improves significantly with respect to the correlation with GDP, and the volatility is close to the data. The correlation between the deposit premium and wages is negative in the data and slightly positive in the model, but the correlation in the data is not statistically different from zero.

The fit of the restricted calibration in which households hold all the deposits is good in some dimensions, but not as good as the full calibration in others. However, if one keeps in mind that various features of the model are designed to produce accurate results in the full calibration with both firms' and households' deposits, then the performance of the restricted calibration appears relatively good.

4.4 Results

To compute the welfare effects of changing capital requirements, we begin by simulating an economy with an 8% capital requirement, a value in between the 7.26% used by Davydiuk (2017) and the 9.25% used by Begenau (2020). We then change the capital requirement to a new level, ζ , and compute the welfare as the economy transitions to the new stochastic steady state.

The solid line in Figure 3 represents welfare in consumption-equivalent percentage units for the full calibration in which both firms and households hold deposits. The optimal capital requirement in this economy is 21.9%. This is much higher than the optimal capital requirement in several models of the literature in which only households hold deposits, such as the 12.4% in Begenau (2020) and the 16% in Begenau and Landvoigt (2021).

The dotted line in Figure 3 represents the welfare in the restricted calibration in which all the deposits are held by households. The optimal capital requirement in this calibration is 14.6%. This is substantially lower than the

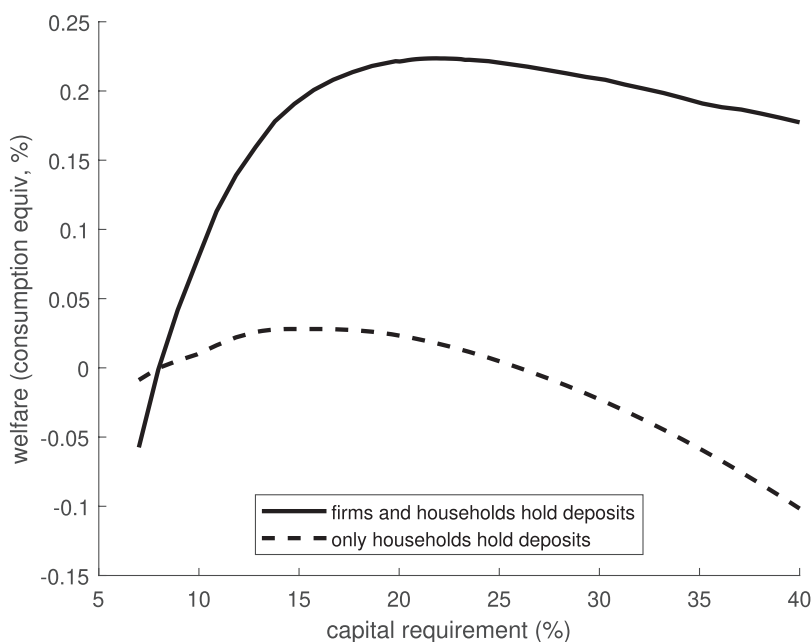


Figure 3
Welfare results

The solid line plots welfare $W(\zeta)$ for the full model calibration in which households and firms hold deposits, and the dotted line plots the welfare for the restricted calibration in which households hold all the deposits. Parameter values are in Table 1. Welfare is defined in consumption-equivalent units. We compute welfare by averaging across 3,000 simulated economies for 300 periods each, where each of the 3,000 simulations is initialized by drawing a point in the state space of the ergodic distribution of the $\zeta = 8\%$ economy.

optimal requirement in the full calibration when firms hold deposits (i.e., 7.3 percentage points lower), though it is in line with the Basel III fully phased-in level of 14%–15% (Basel Committee on Banking Supervision 2017) as well as the optimum in Begenau and Landvoigt (2021) and Begenau (2020).

Figure 3 reveals two key differences between the welfare functions in the two calibrations, in addition to the higher optimum obtained when firms' hold deposits. First, the percent of consumption the household is willing to give up to transition from the $\zeta = 8\%$ economy to the welfare-maximizing economy is an order of magnitude higher in the full calibration (0.224% of annual consumption vs. 0.028%). Second, in the restricted calibration, the welfare function is nearly symmetric around the optimum, a feature shared by other models with only households' deposits, such as Begenau (2020). But in the full calibration that includes firms' deposits, the welfare function is almost flat for capital requirements higher than the optimum, which is again related to firms' deposits and our novel channel. To quantify this difference, we note that the restricted calibration predicts that overshooting the optimal capital requirement by 11.4 percentage points reduces the welfare gains to about zero: households

in the model without firm deposits are indifferent between leaving ζ at 8% and transitioning to $\zeta = 26.0\%$. On the other hand, overshooting the optimal capital requirement by 11.4 percentage points in the full calibration leaves the welfare gain almost unchanged (0.198% when overshooting vs. 0.224% at the optimum). A capital requirement this high in the restricted calibration would actually reduce welfare.

The different shape of the welfare function in the full versus restricted calibrations is a robust feature of our model; it arises in all the alternative calibrations and quantitative exploration exercises we perform in Section 4.5 (see Internet Appendix H). This result is relevant for practical policy implementation. Quantitative models in the capital requirement literature have produced different values for the optimal requirement, and in practice, a regulator might be concerned about the effects of undershooting vs. overshooting the true optimum. Our analysis suggests that it is better for a policymaker who is uncertain about the true optimum to overshoot the optimal requirement rather than undershoot it.

Table 3 shows the impact of a higher capital requirement ζ on the stochastic steady-state in the full calibration. We interpret the values reported in Table 3 as capturing the long-run effects of changes in capital requirements. We have also explored the dynamics of the model as the economy transitions to the new steady state, which we discuss in Internet Appendix H.2.

First, changing capital requirements has large effects on banks' financing (i.e., fewer deposits and more equity), a significant drop in the bank default rate, but only a small reduction in the size of the banking sector (as measured by total bank assets). The lower default rate is mainly driven by a reduction in risk shifting.

Second, firms' long-run demand for deposits is much more price-elastic than households' demand—in a sense, deposits are less important to firms than they are to households. Indeed, the deposit premium increases by only 11 basis points as ζ increases to 21.9%—the same increase in ζ in the restricted calibration with only households' deposits produces an increase of about 30 basis points, despite a similar reduction in deposits. The result can also be seen by noting that firms' deposits drop much more than households' deposits as ζ increases.

Third, higher capital requirements lead to very small changes in real macroeconomic variables. In particular, the apparently small effect on wages does not contradict the intuition outlined in Section 2. The reduction in firms' deposits alone would have large effects on labor. However, firms' labor demand is very elastic and, thus, a small drop in wages is sufficient to prevent a large drop in labor. We also note that (a) the effect of increasing capital requirements on wages in the restricted calibration with only households' deposits is roughly four times smaller than the effect reported here, and (b) the robustness analysis in Section 4.5 further supports the intuition of Section 2 by showing the importance of two key labor market parameters: the Frisch elasticity of labor supply η and the degree of wage rigidities ω . In addition, the improvement in

Table 3
Capital requirements and steady-state values

	Capital requirement			
	8%	15%	21.9%	25%
Bank default (% of all banks)	0.61	0.11	0.02	0.01
Bank default due to risk shifting (% of all banks)	0.53	0.11	0.02	0.01
Bank rental rate of capital, net of depreciation ($r_t - \delta$, %)	4.20	4.24	4.28	4.30
Deposit premium (%)	1.12	1.18	1.23	1.25
Change from $\zeta = 8\%$, percent				
Total deposits	-7.8	-15.3	-18.7	
Households' deposits	-3.79	-6.8	-7.93	
Firms' deposits	-15.9	-32.3	-40.2	
Bank equity	+85.6	+170	+208	
Bank assets	-0.27	-0.29	-0.28	
Output	-0.02	-0.02	-0.01	
Labor	-0.20	-0.25	-0.26	
Capital	-0.27	-0.29	-0.28	
Capital-labor ratio	-0.06	-0.03	-0.003	
Consumption	0.04	0.06	0.05	
Wages	-0.03	-0.04	-0.04	
Return on firm assets R_{t+1}^x , Equation (14)	0.02	0.02	0.02	
Labor share of income	-0.22	-0.29	-0.30	
Capital and profit share of income ($= 1 - \text{labor share}$)	0.51	0.65	0.69	

The table reports average values from the stochastic steady-state distribution, with the value of the capital requirement ζ indicated in the top row. The top panel reports values as percentages, and the bottom panel reports values in percent deviations from the $\zeta = 8\%$ economy.

regulation gives rise to a wealth effect that reduces households' labor supply and counteracts the drop in wages driven by lower labor demand.³⁰

Finally, while our analysis uses a representative household, a concern in practice is that the reduction in wages and labor could have a disproportionate impact on households that derive most of their income from labor. While a full analysis with households' heterogeneity is outside the scope of this paper, Table 3 shows that the labor share of income is hardly affected, dropping by only 0.29%, which is an order of magnitude smaller than the 5 percentage point drop documented by Karabarbounis and Neiman (2014) related to the decline in the price of investment goods.

4.5 Robustness

In this section we illustrate the model mechanisms by exploring alternative parameter values and calibrations. Table 4 reports the results of these robustness exercises, and Figure H2 in Internet Appendix H.2 plots them and provides further discussion. For each exercise, we vary one parameter and/or data target,

³⁰ In particular, our model with separable preferences produces a relatively large impact of the wealth effect on labor supply, in comparison to balanced-growth preferences (King, Plosser, and Rebelo 1988) or Greenwood-Hercowitz-Huffman preferences (Greenwood, Hercowitz, and Huffman 1988; Jaimovich and Rebelo 2009), working against the channel described in Section 2.

and we recalibrate all other parameters to match our target data moments as discussed in Section 4.2.2.³¹

First, we consider a calibration based on a lower liquidity premium on deposits. We follow the approach of some of the papers in the capital requirement literature that assume that bank deposits are close substitutes with Treasury securities (see, e.g., Davydiuk 2017). That is, we compute the liquidity premium on Treasury securities, which is 0.56% in our sample, and use this target for the deposit premium. The 0.56% premium is computed following Krishnamurthy and Vissing-Jorgensen (2012) and adjusting their sample to match ours (see Internet Appendix B.3).³² The calibration matches the lower liquidity premium by changing the parameters ψ and z^L in a way that reduces the demand for deposits, while holding fixed the other targets, in particular, the ratio of households' to firms' deposits. With a lower demand, the cost of raising capital requirements drops and, thus, the optimal capital requirement is higher. Nonetheless, the gap between the full and restricted calibrations remains substantial at almost 3 percentage points.

Next, we consider a simple extension in which firms can invest not only in deposits but also in other liquid assets, such as Treasury securities.³³ With this extension, the optimal capital requirement is 22.5% and thus very close to the 21.9% baseline.³⁴ This is because, when we add other liquid assets, we recalibrate the model to match the other targets and, in particular, the liquidity premium. This is achieved by increasing firms' demand for liquid assets, which offsets the direct effects of the additional supply of liquid assets.

To illustrate our main model mechanism, we solve the model with both a higher and a lower Frisch elasticity: $\eta=1.5$ and $\eta=0.5$. The higher η exceeds the maximum values used in the recent macro-labor literature (Dyrda, Kaplan, and Ríos-Rull 2012), and the lower η is approximately the lower bound implied by Chetty et al. (2011, 2013). The optimal capital requirement gap between the full and restricted calibrations weakens with $\eta=1.5$ but remains large at 3.8 percentage points. For $\eta=0.5$, the gap widens to 10.2 percentage points. These results are in line with the theoretical discussion in Section 2.

We then turn to the degree of wage rigidities. The baseline calibration matches the volatility of wages relative to GDP ($=0.77$), but we also consider alternative measures of wages in the data and find that $[0.73, 0.80]$ represents a reasonable range for this moment (see Internet Appendix B.5). We thus

³¹ In Internet Appendix C.3 and C.4, we perform additional exercises to gauge the effects of the manager contract and firms' idiosyncratic risk on equilibrium outcomes.

³² Using the approach of Davydiuk (2017) delivers approximately the same result. That is, if we measure the liquidity premium as the spread between 3-month AA Commercial Paper and the 3-month Treasury bill, the premium is 59 basis points in our 1986-2010 sample. We thank Tetiana Davydiuk for sharing her data.

³³ We augment the model so that the government issues an amount d^g of Treasury debt. Firms can then invest in d^g so that their total holdings of liquid assets are $\int d_i^l di + d^g$.

³⁴ The optimal capital requirement is unchanged in the restricted calibration in which all the deposits are held by households because we set $d^g=0$ —firms do not hold any liquid assets in that version of the model.

Table 4
Robustness exercises

Calibration	Optimal capital requirement		
	Full (%)	HH only (%)	Difference
Benchmark	21.9	14.6	7.3
Deposit premium = 0.56%	30.8	27.9	2.9
Other liquid assets held by firms = 1.64% of GDP	22.5	14.6	7.9
Frisch elasticity $\eta = 1.5$	21.0	17.2	3.8
Frisch elasticity $\eta = 0.5$	24.4	14.2	10.2
More wage frictions, $\text{vol}(w)/\text{vol}(\text{GDP}) = 0.73$	21.6	19.6	2.0
Less wage frictions, $\text{vol}(w)/\text{vol}(\text{GDP}) = 0.80$	24.6	14.8	9.8
Epstein-Zin preferences, $\text{EIS} = 1.5$	22.5	14.6	7.9
Lower manager risk aversion, $\gamma_m = 2$	22.6	14.6	8.0
Only equity compensation, $\theta_f = \theta_o = 0$	22.1	14.6	7.5
Higher fixed and option compensation, $2\times$	21.9	14.6	7.3
Households' preferences for deposits, $\gamma_d = 0.8$	26.4	20.5	5.9
Deadweight loss from default $\varpi = 4.5\%$	24.5	15.7	8.8
Share of households' deposits 82.8%	24.5	14.6	9.9
No risk shifting, $\lambda_1 = \infty$	3.2	1.9	1.3

The column "Full" indicates the optimal capital requirement in the full calibration with both households' and firms' deposits. The column "HH only" indicates the optimal capital requirement in the restricted calibration in which households hold all the deposits.

conduct two robustness exercises in which we replace the target for the volatility of employment with the lower and upper bound for the volatility of wages, respectively, which we match by varying ω . The difference between the full and restricted calibration changes to 2.0% and 9.8% with more and less wage rigidities, respectively, in line with the discussion in Section 2.

Next, we study the role of managers' preferences and compensation, which affect the optimum in the full calibration but do not play a role in the restricted one without firms' deposits. First, we consider a version of the model in which the manager has Epstein-Zin (EZ) preferences (see Internet Appendix E), setting the additional EZ parameter to obtain an intertemporal elasticity of substitution of 1.5 (Beeler and Campbell 2012). Second, we reduce the risk aversion γ_m of managers to 2 (as in Page 2018). We then change the relative importance of the fixed and option compensation of the manager—first setting them to zero, and then doubling their weight relative to the equity compensation. The optimal capital requirement is nearly unchanged or higher, thereby confirming and strengthening our results.

We then vary γ_d , which affects the elasticity of households' demand for deposits and whose value could affect the optimal regulation (see the discussion in Begenau 2020). We reduce γ_d from 1.4 to 0.8 to illustrate the effect of this change. The optimal capital requirement is higher in both the full and restricted calibrations, but the gap between the two optima remains large (i.e., 5.9 percentage points).

We also extend the model to allow for a deadweight loss from default by assuming that a fraction ϖ of assets b_{t-1} of banks that default at t is lost. We set $\varpi = 4.5\%$ by proxying deadweight losses with the weighted average of total receivership expenses for banks taken over by the FDIC

(Bennett and Unal 2015, table 3). With this additional cost of default, the benefits of raising capital requirements are higher—and more so in the full calibration—thereby strengthening our result and increasing the gap to 8.8 percentage points.

We then conduct an illustrative analysis by increasing the fraction of deposits held by households from 66.4% to 82.8%, which corresponds to the fraction in 1989, and is the highest in our sample (see Internet Appendix B.2). The optimal capital requirement increases from 21.9% to 24.5%, and because the optimum is unchanged in the calibration with only households deposits, the gap that originates from our channel widens to 9.9 percentage points. Though it may appear puzzling that reducing firms' deposits actually increases the strength of a channel that works through firms' deposits, we note that with more households' deposits, we have to reduce the dividend adjustment cost parameter φ to match the volatility of firms' deposits. A lower φ implies that firms' demand for deposits is much more elastic. Thus, as regulators increase capital requirements, the reduction in deposits is absorbed relatively more by firms, and our novel channel plays a bigger role.

Finally, we investigate the role of bad risk-taking by shutting down banks' risk shifting. We do so by setting λ_1 high enough so that banks always choose $p_t = 0$.³⁵ Without risk shifting, the model does not produce the large spikes in banks' default rate that we observe in the data—there are no more banking crises—and thus this exercise substantially reduces the benefit of tightening capital requirements. In this sense, this exercise can be viewed as a way to illustrate the model mechanisms. The optimal capital requirement in the full calibration is 3.2%, which is 1.3 percentage points higher than in the restricted calibration. While this gap is small in absolute value, it remains quite large when compared to the level of the optimal capital requirement in this robustness exercise.

5. Deposits and Good Risk-Taking: Independent Evidence

In this section we provide independent evidence that supports the core mechanism of our model. We have shown that adding firms' deposits to a quantitative dynamic general equilibrium leads to an optimal capital requirement that is higher than in a model in which all deposits are held by households, and that this result is driven by the good risk-taking channel. That is, (a) regulation affects banks' supply of deposits; (b) firms respond by changing their good risk-taking actions; and (c) firms' response is dampened by a general equilibrium effect which operates through wages. The main challenge with providing independent evidence for the good risk-taking channel is to find

³⁵ Setting $\lambda_1 \rightarrow \infty$ implies that there is no need to take a stance on how to calibrate λ_2 (because it only affects banks' choice when they decide to set $p_t > 0$) and v (because $p_t = 0$ implies that the volatility of risk-weighted assets is always the same as that of total assets).

exogenous variation in the supply of deposits. We address this challenge in two ways.

In Section 5.1, we replicate and extend an analysis from Drechsler, Savov, and Schnabl (2017) to show that exogenous changes in deposits affect employment and wages, controlling for the effects that are transmitted directly through bank lending. As in Drechsler, Savov, and Schnabl (2017), we exploit the fact that deposits drop more in counties with higher deposit concentration, following a rate hike by the Federal Reserve.

In a second approach, described in Internet Appendix F, we use the staggered implementation of interstate banking deregulation as an instrument for a change in deposits, following Jayaratne and Strahan (1996), Mian, Sufi, and Verner (2020), and a large literature that has employed this instrument. We find that deregulation triggers an increase in deposits, which in turn has a positive effect on employment and wages. While this analysis represents a less direct way to establish evidence for the good risk-taking channel, it is nonetheless consistent with it, thereby providing additional support.

Overall, these results are consistent with the novel mechanism in our model and support the good risk-taking channel, and they are in line with other studies that have looked directly at the effects of safe asset holdings on firm-level employment, such as Benmelech, Bergman, and Seru (2021).

5.1 The deposit channel of monetary policy and good risk-taking

To provide support for the good risk-taking channel, we build on the approach of Drechsler, Savov, and Schnabl (2017) (henceforth DSS). DSS show that when the Fed raises interest rates, banks widen the spreads they charge on deposits, and deposits flow out of the banking system. Crucially, this effect is stronger in areas in which banks have more market power, which DSS measure using the county-level deposit Herfindahl Hirschman Index (HHI). They label this result the deposit channel of monetary policy.

We build on DSS and use the cross-sectional dispersion in the county-level deposit HHI to generate variation in how deposits respond to changes in monetary policy. The deposit channel of monetary policy predicts that counties in which banks have more market power (i.e., a higher HHI) experience a larger drop of deposits when the Fed funds rate increases. Our objective is to check whether such counties also experience a larger drop in employment and wages, as predicted by the good risk-taking channel. As a crucial step, we can exploit a feature of the deposit channel of DSS to control for the effects on employment and wages that are attributable to changes in bank lending, thus isolating the good risk-taking channel from other channels through which banking shocks affect employment and wages.

We combine the data used by DSS on (a) county-level concentration in the deposit market, (b) Fed funds rates, and (c) county-level employment with (d) county-year-level wages from the March Supplement of the Current Population Survey. We construct an annual panel at the county-year level from 1997 to

Table 5
Deposit channel and good risk-taking

	(1) $\Delta l_{c,t}$	(2) $\Delta l_{c,t}$	(3) $\Delta l_{c,t}$	(4) $\Delta w_{c,t}$	(5) $\Delta w_{c,t}$	(6) $\Delta w_{c,t}$
$\Delta FF_t \times HHI_c$	-0.005*** [0.001]	-0.005*** [0.001]	-0.005*** [0.001]	-0.701** [0.352]	-0.870** [0.349]	-0.673* [0.365]
$\Delta FF_t \times (\text{Lending proxy})_{c,t-1}$			-0.008** [0.003]			-2.170** [1.051]
$(\text{Lending proxy})_{c,t-1}$			0.003 [0.007]			-0.870 [2.862]
Time FE	Y	Y	Y	Y	Y	Y
County FE	Y	Y	Y	Y	Y	Y
County $\times$ ZLB FE	N	Y	Y	N	Y	Y
Observations	54,103	54,103	54,103	54,103	54,103	54,103
R-squared	.147	.202	.202	.108	.163	.163

The data are at the county-year level for the period 1997 to 2013. $\Delta l_{c,t}$ is the logarithmic change in total employment as in DSS. $\Delta w_{c,t}$ is the change in the annual average weekly wage in county c , time t . ΔFF_t is the change in Fed funds rate. HHI_c is the deposit Herfindahl index in county c as in DSS. $(\text{Lending proxy})_{c,t-1}$ corresponds to the variable “County-HHI” in DSS and is constructed as follows: for each bank b , let Bank-HHI_{t-1}^b be the average HHI_c of the counties in which bank b operates, weighted by deposits; then, $(\text{Lending proxy})_{c,t-1}$ is the average Bank-HHI_{t-1}^b for all the banks that operate in county c , weighted by deposits. Standard errors are clustered by county. * $p < .1$; ** $p < .05$; *** $p < .01$.

2013. Our baseline regression is

$$y_{c,t} = \alpha_c + \alpha_t + \beta \Delta FF_t \times HHI_c + \varepsilon_{c,t}, \tag{33}$$

where $y_{c,t}$ is the logarithm of the growth in employment or the change in wages in county c and time t , FF_t is the Fed funds rate at time t , and HHI_c is the deposit Herfindahl index that measures concentration in the deposit market in county c . We construct HHI_c as described in DSS. Table 5 presents the estimate of our baseline regression. Columns 1-3 contain the results about employment, and columns 4-6 those about wages.

Column 1 of Table 5 shows that fed funds shocks reduce employment more in counties with higher deposit concentration; column 4 shows that wages also respond more to fed funds shocks in counties with higher deposit concentration. These two results, combined with the central insight of DSS, suggest that a reduction in deposits reduces labor demand, consistent with the good risk-taking channel. Columns 2 and 5 add county fixed effects interacted with a dummy for the zero-lower-bound period, following DSS, and the results are essentially unchanged.

The results in columns 1-2 and 4-5 of Table 5 are consistent with the good risk-taking channel, but they also may be driven by a standard bank-lending channel. That is, fed funds shocks might reduce aggregate lending demand overall, generating a reduction in labor demand by firms. To control for the effects of bank lending, we build on another strategy developed by DSS, who conjecture that counties with banks that experience larger outflows of deposits due to the deposit channel—and that do so not just in county c but *across all the counties they operate*—reduce new lending (table VII, columns 1 and 2 in DSS). They show that this is indeed the case by constructing a proxy for the

deposit outflow; columns 3 and 6 include the same proxy in our regression and show that our results are robust to this control.

Although column 3 of Table 5 is the same as one of the regressions in DSS (in particular, the same as table VII, column 4 in DSS), our purpose in estimating it is different.³⁶ DSS are interested in studying how the deposit channel affects the labor market through lending, controlling for the direct effects of deposits. For us, it is the opposite: we want to understand the effects of deposits on the labor market, controlling for lending. To that end, the additional results in columns 4–6 that the reduction in deposits also reduces wages provides additional evidence in favor of the good risk-taking channel.

The results presented in this section are useful in providing external support for the good risk-taking channel, but we cannot incorporate them quantitatively into our model. This is because the model in Section 3 is primarily concerned with generating results in general equilibrium, while the results in Table 5 are only identified in the cross-section of counties, controlling for aggregate general equilibrium effects. While it would be an interesting exercise to extend our model to include heterogeneous geographic areas with different degrees of bank monopoly power and examine how this heterogeneity interacts with the good risk-taking channel, that is far beyond the scope of the present paper.

6. Conclusion

We study capital requirement regulation when both households and firms hold deposits. Because of a novel general equilibrium effect related to firms' deposits and wages, the optimal capital requirement is substantially higher than in comparable models—used in most of the literature—in which all the deposits are held by households.

To keep our model tractable and comparable to most of the literature on optimal capital requirements, we assumed that all households are alike and all banks are alike. We acknowledge, however, that departing from these modeling assumptions would give rise to additional policy implications. For instance, households in our model are both workers and owners of the firms; in a model in which workers and owners are separate agents, a change in capital requirements that alters wages and firms' profits might give rise to policy-relevant redistributive effects. In addition, all the banks in our model cater equally to firms and households; if banks instead specialize in serving particular agents, we conjecture that it might be optimal to impose a looser capital requirement on banks that serve households and a tighter one on those that cater more to firms' deposits.

This paper opens up several directions for future research. First, depending on how firms' idiosyncratic productivities evolve over time, the value of deposits

³⁶ The number of observation is slightly smaller for us because we are missing wage data for a few county-year observations.

to firms might be time varying. This effect might give rise to important implications for the analysis of time-varying regulation above and beyond those analyzed by Davydiuk (2017) and Malherbe (2020), in which the value of deposits is constant over time and time variation in optimal capital requirements comes from time variation in banks' overinvestment. Second, we have followed the literature in assuming complete deposit insurance, but Egan, Hortaçsu, and Matvos (2017) note that only about half of all deposits in the United States are in fact FDIC insured. Our model can be employed to study the optimal degree of deposit insurance. In our model, eliminating deposit insurance altogether increases firms' cash flow volatility, and thus the optimal outcome is likely to be partial deposit insurance.

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